

13.1 Planes and Surfaces

- Find equations of the plane, given a normal vector and a point; given a set of points.
ex. $P_0(1, 0, -3)$; $\mathbf{n} = \langle 1, -1, 2 \rangle$
ex. $(2, -1, 4)$, $(1, 1, -1)$, and $(-1, 3, 1)$

13.2 Planes and Surfaces

- Find the domain and range of a given function.
ex. $f(x, y) = \sin\left(\frac{x}{y}\right)$, $f(x, y) = \sqrt{x - 2y + 4}$
- Be able to read level curves.

13.3 Limits and Continuity

- Limits of two-variable continuous functions.
ex. $\lim_{(x,y) \rightarrow (2,0)} (xy^5 - x)$, $\lim_{(x,y) \rightarrow (e^2, 4)} \ln \sqrt{xy}$
- Limits at the boundary points of two-variable functions.
ex. $\lim_{(x,y) \rightarrow (6,2)} \frac{x^2 - 3xy}{y - 3y}$, $\lim_{(x,y) \rightarrow (1,2)} \frac{\sqrt{y} - \sqrt{x+1}}{y - x - 1}$
- Nonexistence of limits using the Two-Path Test.
ex. $\lim_{(x,y) \rightarrow (0,0)} \frac{x + 2y}{x - 2y}$, use lines $x = 0$ and then $y = 0$.
ex. $\lim_{(x,y) \rightarrow (0,0)} \frac{y^3 + x^3}{xy^2}$, use lines $x = y$ and then $x = -y$.

13.4 Partial Derivatives

- Given functions $f(x, y)$, find f_x , f_y , f_{xx} , f_{yy} , f_{xy} , and f_{yx} .
ex. $f(x, y) = e^{x+7}$, $\cos xy$, $2x^3 + 3y^2 + 1$

13.5 The Chain Rule

- Use the Chain rule to find the derivatives of functions with one independent variable (only two-variable functions).
ex. Find dz/dt where $z = x \sin y$, $x = t^2$, and $y = 4t^3$

13.6 Directional Derivatives and the Gradient

- Compute the gradient of two-variable functions.
ex. Let $f(x, y) = 4x^2 - 2xy + y^2$. Find $\nabla f(x, y)$ at $P(-1, -5)$. Equivalently, find the steepest ascent of $f(x, y)$ at $P(-1, -5)$.
- Find a unit vector that points in the direction of the steepest ascent and descent of a function.
- Find a vector that gives a direction of no change of a function.
ex. For $f(x, y) = x^2 - 4y^2 - 9$ at $P(1, -2)$

13.8 Maximum and Minimum Problems

- Find the critical points of a function, and use the Second Derivative Test to determine (when possible) whether each critical point corresponds to a local maximum, minimum, or saddle point.
ex. $f(x, y) = -4x^2 + 8y^2 - 3$, $f(x, y) = \sqrt{x^2 + y^2 - 4x + 5}$, $f(x, y) = ye^x - e^y$.